$$G = \nabla f(W)$$

$$X_0 = G/(\|G\|_F + \varepsilon)$$

repeat $k = 0 \ldots K-1$:

$$A = X_k X_k^T$$

$$B = 1.5A - 0.5A^2$$

$$X_{k+1} = BX_k$$

$$X_K \approx \text{orthogonal direction}$$

$$W \leftarrow W - \eta X_K$$